# Macro & Market Impact Analytics API

1.0.0 OAS 3.1

/openapi.json

A REST API for tracking macroeconomic indicators and financial market assets, with analytical endpoints for correlation analysis, recession risk scoring, sector impact analysis, and macro trend detection.

## Indicators

## Assets

GET `/api/v1/assets` List all assets

Returns all tracked market assets. Optionally filter by sector, asset class, or country.

**Parameters** Try it out

| Name | Description |
|---|---|
| sector *(query)* | Filter by sector (e.g. Technology)<br>sector |
| asset_class *(query)* | Filter by asset class<br>asset_class |
| country *(query)* | Filter by country (e.g. US)<br>country |
| skip integer *(query)* minimum: 0 | Number of records to skip<br>*Default value* : 0<br>0 |
| limit integer *(query)* maximum: 500 minimum: 1 | Maximum records to return<br>*Default value* : 100<br>100 |

### Responses

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br>Media type<br>**application/json**<br>Controls `Accept` header.<br>**Example Value** Schema | *No links* |
| 422 | Validation Error<br>Media type<br>**application/json**<br>**Example Value** Schema | *No links* |

```
[
  {
    "ticker": "SPY",
    "name": "SPDR S&P 500 ETF Trust",
    "asset_class": "etf",
    "sector": "Broad Market",
    "country": "US",
    "description": "Tracks the S&P 500 index",
    "id": 0,
    "created_at": "2026-03-20T00:31:39.776Z",
    "updated_at": "2026-03-20T00:31:39.776Z"
  }
]
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

**POST** `/api/v1/assets` Create a new asset

Adds a new market asset to the database. The ticker symbol must be unique.

### Parameters

Try it out

No parameters

**Request body** required

application/json

**Example Value** Schema

```
{
  "ticker": "SPY",
  "name": "SPDR S&P 500 ETF Trust",
  "asset_class": "etf",
  "sector": "Broad Market",
  "country": "US",
  "description": "Tracks the S&P 500 index"
}
```

**Responses**

| Code | Description | Links |
|---|---|---|
| 201 | Successful Response<br>Media type<br>application/json<br>Controls `Accept` header.<br>**Example Value** Schema | *No links* |
| 409 | Asset with this ticker already exists<br>**Example Value** | *No links* |
| 422 | Validation Error<br>Media type<br>application/json<br>**Example Value** Schema | *No links* |

```
{
  "ticker": "SPY",
  "name": "SPDR S&P 500 ETF Trust",
  "asset_class": "etf",
  "sector": "Broad Market",
  "country": "US",
  "description": "Tracks the S&P 500 index",
  "id": 0,
  "created_at": "2026-03-20T00:31:39.785Z",
  "updated_at": "2026-03-20T00:31:39.785Z"
}
```

```
"string"
```

| Code | Description | Links |
|---|---|---|


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


GET `/api/v1/assets/{asset_id}` Get asset by ID

Returns a single market asset with its full OHLCV price snapshot history.

**Parameters**

Try it out

| Name | Description |
|---|---|
| **asset_id** * required<br>`integer`<br>*(path)* | asset_id |

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br><br>Media type<br>**application/json**<br>Controls `Accept` header.<br><br>**Example Value** Schema | *No links* |


```
{
  "ticker": "SPY",
  "name": "SPDR S&P 500 ETF Trust",
  "asset_class": "etf",
  "sector": "Broad Market",
  "country": "US",
  "description": "Tracks the S&P 500 index",
  "id": 0,
  "created_at": "2026-03-20T00:31:39.790Z",
  "updated_at": "2026-03-20T00:31:39.790Z",
```

| Code | Description | Links |
|---|---|---|

```
  "snapshots": []
}
```

404 Asset not found — *No links*

**Example Value**

```
"string"
```

422 Validation Error — *No links*

Media type

**application/json**

**Example Value** Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

**PATCH** `/api/v1/assets/{asset_id}` Update an asset

Partially updates an existing market asset. Only provided fields are updated.

**Parameters** **Try it out**

| Name | Description |
|---|---|
| **asset_id** * required<br>`integer`<br>*(path)* | asset_id |

**Request body** required **application/json**

**Example Value** Schema

```
{
  "name": "string",
  "asset_class": "equity",
  "sector": "string",
```

```
  "country": "string",
  "description": "string"
}
```

## Responses

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br><br>Media type<br>**application/json**<br>Controls `Accept` header.<br><br>**Example Value** Schema | *No links* |
| 404 | Asset not found<br><br>**Example Value** | *No links* |
| 422 | Validation Error<br><br>Media type<br>**application/json**<br><br>**Example Value** Schema | *No links* |

```
{
  "ticker": "SPY",
  "name": "SPDR S&P 500 ETF Trust",
  "asset_class": "etf",
  "sector": "Broad Market",
  "country": "US",
  "description": "Tracks the S&P 500 index",
  "id": 0,
  "created_at": "2026-03-20T00:31:39.800Z",
  "updated_at": "2026-03-20T00:31:39.800Z"
}
```

```
"string"
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

DELETE `/api/v1/assets/{asset_id}` Delete an asset

Permanently deletes an asset and all its associated price snapshots (cascade delete).

**Parameters**

Try it out

| Name | Description |
|---|---|
| **asset_id** * required<br>`integer`<br>*(path)* | asset_id |

**Responses**

| Code | Description | Links |
|---|---|---|
| 204 | Successful Response<br>**Example Value**<br>`"string"` | *No links* |
| 404 | Asset not found<br>**Example Value**<br>`"string"` | *No links* |
| 422 | Validation Error<br>Media type<br>**application/json**<br>**Example Value** Schema | *No links* |

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
```

| Code | Description | Links |
|---|---|---|

```
    }
  ]
}
```

# Snapshots & Ingestion

POST `/api/v1/snapshots/indicators/{indicator_id}/ingest` Ingest indicator data from FRED

Pulls the full observation history for the given indicator from the FRED API and stores any new observations in the database. Safe to call repeatedly — already-existing dates are skipped automatically.

**Parameters** Try it out

| Name | Description |
|---|---|
| **indicator_id** * required<br>`integer`<br>*(path)* | indicator_id |

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br>Media type<br>**application/json**<br>Controls `Accept` header.<br>**Example Value** Schema | *No links* |
| 404 | Indicator not found<br>**Example Value** | *No links* |

```
{
  "indicator_id": 0,
  "fred_series_id": "string",
  "inserted": 0,
  "skipped": 0,
  "message": "string"
}
```

```
"string"
```

| Code | Description | Links |
|---|---|---|
| 422 | FRED API key not configured or invalid series ID<br>**Example Value**<br>`"string"` | *No links* |

GET `/api/v1/snapshots/indicators/{indicator_id}` Get indicator snapshots

Returns time-series observations for an indicator, with optional date range filtering.

**Parameters** Try it out

| Name | Description |
|---|---|
| **indicator_id** * required<br>`integer`<br>*(path)* | indicator_id |
| start_date<br>*(query)* | Filter from this date (inclusive)<br>start_date |
| end_date<br>*(query)* | Filter to this date (inclusive)<br>end_date |
| limit<br>`integer`<br>*(query)*<br>*maximum: 5000*<br>*minimum: 1* | Maximum records to return<br>*Default value* : 500<br>500 |

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br>Media type | *No links* |

| Code | Description | Links |
|---|---|---|

**application/json**

Controls `Accept` header.

**Example Value** Schema


```
[
  {
    "date": "2026-03-20",
    "value": 0,
    "id": 0,
    "indicator_id": 0,
    "created_at": "2026-03-20T00:31:39.817Z"
  }
]
```


404 Indicator not found *No links*

**Example Value**


```
"string"
```


422 Validation Error *No links*

Media type

**application/json**

**Example Value** Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


POST `/api/v1/snapshots/assets/{asset_id}/ingest` Ingest asset price data from Yahoo Finance

Pulls the full OHLCV price history for the given asset from Yahoo Finance and stores any new records in the database. Daily returns are computed automatically. Safe to call repeatedly — already-existing dates are skipped.

**Parameters** Try it out

| Name | Description |
| --- | --- |
| **asset_id** * required<br>integer<br>*(path)* | asset_id |

### Responses

| Code | Description | Links |
| --- | --- | --- |
| 200 | Successful Response<br><br>Media type<br>**application/json**<br>Controls `Accept` header.<br><br>**Example Value** Schema<br><br>`{ "asset_id": 0, "ticker": "string", "inserted": 0, "skipped": 0, "message": "string" }` | *No links* |
| 404 | Asset not found<br><br>**Example Value**<br><br>`"string"` | *No links* |
| 422 | Invalid ticker or no data returned<br><br>**Example Value**<br><br>`"string"` | *No links* |

GET `/api/v1/snapshots/assets/{asset_id}` Get asset price snapshots

Returns OHLCV price history for an asset, with optional date range filtering.

**Parameters** Try it out

| Name | Description |
|---|---|
| **asset_id** * required<br>integer<br>*(path)* | asset_id |
| start_date<br>*(query)* | Filter from this date (inclusive)<br>start_date |
| end_date<br>*(query)* | Filter to this date (inclusive)<br>end_date |
| limit<br>integer<br>*(query)*<br>*maximum: 5000*<br>*minimum: 1* | Maximum records to return<br>*Default value* : 500<br>500 |

## Responses

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br><br>Media type<br>**application/json**<br>Controls `Accept` header.<br><br>**Example Value** Schema | *No links* |
| 404 | Asset not found<br><br>**Example Value** | *No links* |

```
[
  {
    "date": "2026-03-20",
    "open": 0,
    "high": 0,
    "low": 0,
    "close": 0,
    "volume": 0,
    "daily_return": 0,
    "id": 0,
    "asset_id": 0,
    "created_at": "2026-03-20T00:31:39.830Z"
  }
]
```

```
"string"
```

| Code | Description | Links |
|---|---|---|
| 422 | Validation Error<br>Media type<br>application/json<br>**Example Value** Schema | *No links* |

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

# Analytics

GET `/api/v1/analytics/correlation` Correlation analysis

Computes Pearson correlation between a macroeconomic indicator and a market asset. Both series are resampled to monthly frequency before alignment. Requires data to have been ingested for both the indicator and asset.

**Parameters** Try it out

| Name | Description |
|---|---|
| **indicator_id** * required<br>`integer`<br>*(query)* | ID of the macroeconomic indicator<br>indicator_id |
| **asset_id** * required<br>`integer`<br>*(query)* | ID of the market asset<br>asset_id |
| start_date<br>*(query)* | Start of analysis window<br>start_date |
| end_date<br>*(query)* | End of analysis window<br>end_date |

## Responses

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br>Media type<br>**application/json**<br>Controls `Accept` header.<br>**Example Value** Schema | *No links* |
| 404 | Indicator or asset not found<br>**Example Value** | *No links* |
| 422 | Insufficient data for analysis<br>**Example Value** | *No links* |

```
{
  "indicator_id": 0,
  "indicator_name": "string",
  "asset_id": 0,
  "asset_ticker": "string",
  "correlation": 0,
  "interpretation": "string",
  "data_points": 0,
  "start_date": "2026-03-20",
  "end_date": "2026-03-20",
  "series": [
    {
      "date": "2026-03-20",
      "indicator_value": 0,
      "asset_value": 0
    }
  ]
}
```

```
"string"
```

```
"string"
```

GET `/api/v1/analytics/recession-risk` Recession risk score

Computes a composite recession risk score (0–100) from four established macroeconomic signals: yield curve spread (T10Y2Y), unemployment rate (UNRATE), CPI inflation (CPIAUCSL), and the federal funds rate (FEDFUNDS). Signals are weighted by their relative predictive power in the academic literature. Ingest all four indicators for the most accurate score.

**Parameters** Try it out

No parameters

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br>Media type<br>application/json<br>Controls `Accept` header.<br>**Example Value** Schema | *No links* |


```
{
  "score": 0,
  "level": "string",
  "color": "string",
  "summary": "string",
  "signals": [
    {
      "name": "string",
      "fred_series_id": "string",
      "current_value": 0,
      "signal_score": 0,
      "weight": 0,
      "contribution": 0,
      "interpretation": "string"
    }
  ],
  "as_of_date": "2026-03-20"
}
```


GET `/api/v1/analytics/macro-trend/{indicator_id}` Macro trend detection

Fits a linear regression to the last N observations of a macro indicator to determine trend direction (rising/falling/stable) and percentage change. The slope is normalised relative to the mean value for cross-indicator comparability.

**Parameters** Try it out

| Name | Description |
|---|---|
| **indicator_id** * required<br>`integer`<br>*(path)* | indicator_id |

| Name | Description |
|---|---|
| periods<br>integer<br>*(query)*<br>*maximum: 120*<br>*minimum: 3* | Number of monthly periods to analyse<br>*Default value* : 24<br>24 |

### Responses

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br>Media type<br>**application/json**<br>Controls `Accept` header.<br>**Example Value** Schema | *No links* |
| 404 | Indicator not found<br>**Example Value** | *No links* |
| 422 | Insufficient data for the requested number of periods<br>**Example Value** | *No links* |

```
{
  "indicator_id": 0,
  "indicator_name": "string",
  "fred_series_id": "string",
  "periods": 0,
  "direction": "string",
  "change_pct": 0,
  "latest_value": 0,
  "earliest_value": 0,
  "latest_date": "2026-03-20",
  "earliest_date": "2026-03-20",
  "slope": 0,
  "interpretation": "string"
}
```

```
"string"
```

```
"string"
```

GET `/api/v1/analytics/sector-impact/{indicator_id}` Sector impact analysis

Correlates a macro indicator against every tracked market asset, returning results sorted by absolute correlation strength. Assets with insufficient aligned data are excluded and counted separately. Use this to identify which sectors are most exposed to a given macro variable.

## Parameters

Try it out

| Name | Description |
|---|---|
| **indicator_id** * required<br>`integer`<br>*(path)* | indicator_id |

## Responses

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br><br>Media type<br>**application/json**<br>Controls `Accept` header. | *No links* |

**Example Value** Schema


```
{
  "indicator_id": 0,
  "indicator_name": "string",
  "fred_series_id": "string",
  "results": [
    {
      "asset_id": 0,
      "ticker": "string",
      "asset_name": "string",
      "sector": "string",
      "asset_class": "string",
      "correlation": 0,
      "data_points": 0,
      "interpretation": "string"
    }
  ],
  "total_assets_analysed": 0,
  "insufficient_data_assets": 0
}
```

| Code | Description | Links |
|---|---|---|
| 422 | No indicator data available<br>**Example Value**<br>`"string"` | *No links* |

GET `/api/v1/analytics/market-summary` Market summary

Returns the latest value, trend direction, and period-over-period change for every tracked indicator and asset. Designed to power the dashboard home page. Assets and indicators with no ingested data are included with null values.

**Parameters** Try it out

No parameters

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br>Media type<br>**application/json**<br>Controls `Accept` header.<br>**Example Value** Schema | *No links* |


```
{
  "indicators": [
    {
      "id": 0,
      "fred_series_id": "string",
      "name": "string",
      "category": "string",
      "latest_value": 0,
      "latest_date": "2026-03-20",
      "previous_value": 0,
      "change": 0,
      "change_pct": 0,
      "trend": "string"
    }
  ],
  "assets": [
    {
      "id": 0,
      "ticker": "string",
```

| Code | Description | Links |
|---|---|---|


```
    "name": "string",
    "asset_class": "string",
    "sector": "string",
    "latest_close": 0,
    "latest_date": "2026-03-20",
    "daily_return": 0,
    "trend": "string"
  }
```


# Health

GET /health Health check

## Schemas

### AssetClass

Collapse all string

Allowed values "equity" "etf" "index" "commodity" "bond" "other"

### AssetCreate

Collapse all object

ticker* string <= 20 characters

name* string <= 200 characters

asset_class* Expand all string

sector Expand all (string | null)

country Expand all (string | null)

description Expand all (string | null)

### AssetIngestResponse

Collapse all object

asset_id* integer

ticker* string

inserted* integer

skipped* integer

message* string

### AssetResponse

Collapse all object

ticker* string <= 20 characters

name* string <= 200 characters

asset_class* Expand all string

sector Expand all (string | null)

country > Expand all (string | null)

description > Expand all (string | null)

id* integer

created_at* string date-time

updated_at* string date-time

### AssetSnapshotResponse ^ Collapse all object

date* string date

open > Expand all (number | null)

high > Expand all (number | null)

low > Expand all (number | null)

close* number

volume > Expand all (integer | null)

daily_return > Expand all (number | null)

id* integer

asset_id* integer

created_at* string date-time

### AssetSummary ^ Collapse all object

id* integer

ticker* string

name* string

asset_class* string

sector* > Expand all (string | null)

latest_close* > Expand all (number | null)

latest_date* > Expand all (string | null)

daily_return* > Expand all (number | null)

trend* string

### AssetUpdate ^ Collapse all object

name > Expand all (string | null)

asset_class > Expand all (string | null)

sector > Expand all (string | null)

country > Expand all (string | null)

description > Expand all (string | null)

### AssetWithSnapshots ^ Collapse all object

ticker* string <= 20 characters

name* string <= 200 characters

asset_class* > Expand all string

sector > Expand all (string | null)

country > Expand all (string | null)

description > Expand all (string | null)

id* integer

created_at* string date-time

updated_at* string date-time

snapshots > Expand all array<object>

## CorrelationPoint ^ Collapse all object

A single aligned data point used in correlation analysis.

date* string date

indicator_value* number

asset_value* number

## CorrelationResponse > Expand all object

## HTTPValidationError ^ Collapse all object

detail > Expand all array<object>

## IndicatorCategory ^ Collapse all string

**Allowed values**

"inflation" "interest_rate" "gdp" "unemployment" "yield_curve" "credit" "other"

## IndicatorCreate ^ Collapse all object

fred_series_id* string <= 50 characters

name* string <= 200 characters

description > Expand all (string | null)

category* > Expand all string

unit > Expand all (string | null)

frequency > Expand all (string | null)

source > Expand all string <= 100 characters

## IndicatorResponse ^ Collapse all object

fred_series_id* string <= 50 characters

name* string <= 200 characters

description > Expand all (string | null)

category* > Expand all string

unit > Expand all (string | null)

frequency > Expand all (string | null)

source > Expand all string <= 100 characters

id* integer

created_at* string date-time

updated_at* string date-time

### IndicatorSummary ^ Collapse all object

id* integer

fred_series_id* string

name* string

category* string

latest_value* > Expand all (number | null)

latest_date* > Expand all (string | null)

previous_value* > Expand all (number | null)

change* > Expand all (number | null)

change_pct* > Expand all (number | null)

trend* string

### IndicatorUpdate ^ Collapse all object

name > Expand all (string | null)

description > Expand all (string | null)

category > Expand all (string | null)

unit > Expand all (string | null)

frequency > Expand all (string | null)

source > Expand all (string | null)

### IndicatorWithSnapshots ^ Collapse all object

fred_series_id* string <= 50 characters

name* string <= 200 characters

description > Expand all (string | null)

category* > Expand all string

unit > Expand all (string | null)

frequency > Expand all (string | null)

source > Expand all string <= 100 characters

id* integer

created_at* string date-time

updated_at* string date-time

snapshots > Expand all array<object>

### IngestResponse ^ Collapse all object

indicator_id* integer

fred_series_id* string

inserted* integer

skipped* integer

message* string

### MacroTrendResponse ^ Collapse all object

indicator_id* integer

indicator_name* string

fred_series_id* string

periods* integer

direction* string

change_pct* number

latest_value* > Expand all (number | null)

earliest_value* > Expand all (number | null)

latest_date* > Expand all (string | null)

earliest_date* > Expand all (string | null)

slope* number

interpretation* string

### MarketSummaryResponse ^ Collapse all object

indicators* > Expand all array<object>

assets* > Expand all array<object>

as_of* string

### RecessionRiskResponse ^ Collapse all object

score* number

level* string

color* string

summary* string

signals* > Expand all array<object>

as_of_date* > Expand all (string | null)

### RecessionSignal ^ Collapse all object

Individual signal contribution to the composite recession risk score.

name* string

fred_series_id* string

current_value* > Expand all (number | null)

signal_score* number

weight* number

contribution* number

interpretation* string

### SectorCorrelation ^ Collapse all object

asset_id* integer

ticker* string

asset_name* string

sector* > Expand all (string | null)

asset_class* string

correlation* number

data_points* integer

interpretation* string

### SectorImpactResponse ^ Collapse all object

indicator_id* integer

indicator_name* string

fred_series_id* string

results* > Expand all array<object>

total_assets_analysed* integer

insufficient_data_assets* integer

### SnapshotResponse ^ Collapse all object

date* string date

value* number

id* integer

indicator_id* integer

created_at* string date-time

## ValidationError ^ Collapse all `object`

`loc*` > Expand all `array<(string | integer)>`

`msg*` `string`

`type*` `string`